Adnan Husain

SENIOR SOFTWARE ENGINEER | FOUNDER

Calgary, Alberta

LinkedIn | github.com/adnanrh | Research portfolio | Technical writing

Backend-focused full-stack engineer and founder with 5+ years of production experience across AWS S3, Cisco Secure Endpoint, and independent product development. I build reliable systems from architecture and APIs through infrastructure, testing, and delivery. I publish AI-assisted mathematical investigations, Python verification tools, and technical writing.

PROFESSIONAL EXPERIENCE

Meteor City

2026 - Present

Founder & Software Engineer | Calgary, AB | Private pre-launch development

- Architected and built an invitation-only, privacy-first social platform with Rails 8 and PostgreSQL, including identity verification, reusable invitations, mutual connections, and privacy-scoped access controls.
- Directed architecture, implementation, testing, and deployment using ChatGPT and Codex, retaining explicit human review and acceptance gates.
- Implemented private image/video uploads and a chronological feed with stable publication ordering, explicit pagination, and user-controlled caught-up checkpoints.
- Built staging infrastructure with Docker, Kamal, DigitalOcean, Cloudflare R2, Twilio Verify, Resend, and background processing; verified authorization and failure handling through automated and live tests.

Amazon Web Services (S3)

Mar 2024 - Feb 2025

Software Development Engineer II | Vancouver, BC

- Contributed to S3 Index Control Plane teams maintaining the distributed key-value system that mapped S3 keys to objects and metadata.
- Increased scaling runway by **50%** in high-load S3 regions by balancing key ranges across Index cells while preserving low-latency service objectives.
- Designed a node-capacity candidate-selection algorithm with projected **10% annual savings** on S3 Index infrastructure costs.

Cisco Systems

Calgary, AB

Senior Software Engineer — Oct 2022 - Aug 2023

- Designed and implemented a REST API automating Secure Endpoint Connector releases, reducing release-related engineering effort by **80%**.
- Upgraded an **80,000+ line** Rails codebase across three major/minor versions; created a repeatable runbook that reduced end-to-end upgrade rollout time by **60%**.

Software Engineer — Oct 2019 - Oct 2022

- Integrated sampling-based application performance monitoring and optimized the 10 slowest endpoints in Cisco Secure Endpoint, reducing average response time by **40%**.
- Introduced CI checks for N+1 queries and Rails antipatterns, reducing latency and HTTP 5xx response rates by **50%** during release windows.

INDEPENDENT RESEARCH & TECHNICAL WRITING

Recursive Prompting Experiments

2026

Project Creator & Research Facilitator | AI-assisted mathematical investigation

github.com/adnanrh/recursive-prompting-experiments

Created a three-part exploration of large language model (LLM) reasoning in number theory, graph theory, and computational complexity. Defined the goals and facilitated a workflow in which an AI model wrote its own research prompt, then received that prompt for execution.

- Published standard-library Python verification tools and supporting artifacts alongside technical writing documenting results, failed approaches, and verification limits.

001 Erdős–Straus conjecture Read article

- Published an exact-integer-arithmetic audit covering all **9,592 primes below 100,000**, with cross-checks between factor-pair and reduced-fraction methods and recorded computational certificates. Classified the outcome as a reproduction of existing results.

002 Graceful Tree conjecture Read article

- Documented constructive approaches and counterexample searches; published a JSON-certificate verifier checking valid tree structure, bijective vertex labels, and complete, distinct edge differences, with explicit rejection of malformed inputs.

003 P versus NP Read article

- Published verification tooling for **five finite pigeonhole-principle certificates** as part of an exploration of proof complexity, with checks for Boolean formula encodings, SHA-256 integrity, and algebraic identity and dual-certificate validation.

Scope: exploratory, AI-assisted research. Finite computational checks are distinguished from general proofs; no resolution of the open problems is claimed.

TECHNICAL EXPERTISE

Languages & web Ruby, Python, Java, Kotlin, Scala, JavaScript; Rails, Vue.js, REST APIs

Data & systems PostgreSQL, NoSQL, distributed storage, concurrency, transaction design, performance optimization

Infrastructure AWS S3, Docker, Kamal, DigitalOcean, Cloudflare R2, CI/CD, background jobs

Engineering & research Automated testing, access control, AI-assisted development (ChatGPT / Codex), experiment design, computational verification, technical writing

EDUCATION

University of Calgary Sep 2013 - Apr 2019

BSc in Software Engineering | 3.5/4.0 GPA | Dean's List (Years 1 and 2)

The University of Hong Kong Sep 2015 - May 2016

Exchange studies contributing toward an Applied Mathematics minor